

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Report

Executive Summary

This report evaluates `z-ai/glm-5.2` across Infron and OpenRouter in prompt-caching workloads. The experiment uses only `/v1/chat/completions`, keeps model/platform default reasoning and thinking behavior, and keeps prompt-length stratification enabled.

After A/B filtering, the analysis retains **757 paired samples** and **3028 request-level observations**. It excludes **86 records**.

Core finding: Infron leads observed cost in every routing mode. OpenRouter leads throughput, E2E latency, and Streaming TTFT in every routing mode. Cache-rate winners are split: Infron leads Throughput First and Price First, while OpenRouter slightly leads Latency First and TTFT First.

Quality Gate

Item	Value
Model	<code>z-ai/glm-5.2</code>
API protocol	<code>/v1/chat/completions</code> only
Groups / rounds	4 groups x 50 rounds
Prompt tiers	<code>short</code> about 1500, <code>medium</code> about 8000, <code>long</code> about 32000 tokens
Reasoning / thinking	Default platform/model behavior; no explicit disable parameter
Pairing rule	strict sort/group/round pair with first/second usage.prompt_tokens deltas <= 50
Retained pairs	757
Request-level rows	3028
Excluded records	86

Route-Level Result Matrix

Routing mode	Pairs	Token cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Throughput First	186	Infron 94.17% / OpenRouter 58.11%	Infron \$1.301319 / OpenRouter \$4.617887	Infron 1.756 tok/s / OpenRouter 3.028 tok/s	Infron 7726.33 ms / OpenRouter 5284.79 ms	Infron 7082.53 ms / OpenRouter 4977.95 ms

Routing mode	Pairs	Token cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Price First	190	Infron 99.58% / OpenRouter 90.96%	Infron \$0.878579 / OpenRouter \$1.521277	Infron 2.029 tok/s / OpenRouter 2.346 tok/s	Infron 6974.85 ms / OpenRouter 6821.29 ms	Infron 6378.09 ms / OpenRouter 6331.30 ms
Latency First	189	Infron 99.58% / OpenRouter 99.83%	Infron \$0.868005 / OpenRouter \$1.827801	Infron 1.884 tok/s / OpenRouter 5.129 tok/s	Infron 7395.10 ms / OpenRouter 3119.52 ms	Infron 6710.94 ms / OpenRouter 2709.15 ms
TTFT First	192	Infron 99.62% / OpenRouter 99.82%	Infron \$0.872405 / OpenRouter \$1.625334	Infron 2.976 tok/s / OpenRouter 4.892 tok/s	Infron 4644.30 ms / OpenRouter 3270.76 ms	Infron 4023.05 ms / OpenRouter 2866.14 ms

API Protocol Record

API protocol	Endpoint	Note
chat_completions	/v1/chat/completions	This run keeps only the standard Chat Completions protocol; it does not include /v1/messages or /v1/responses .

Prompt–Length Stratified Cache Performance

Tier	Pairs	Token cache hit rate	Call cache hit rate	E2E latency	TTFT	Reasoning tokens
short	259	Infron 91.86% / OpenRouter 84.29%	Infron 97.30% / OpenRouter 88.80%	Infron 5094.35 ms / OpenRouter 3443.73 ms	Infron 4437.02 ms / OpenRouter 3031.17 ms	Infron 0 / OpenRouter 8845
medium	251	Infron 98.23% / OpenRouter 86.34%	Infron 98.41% / OpenRouter 88.84%	Infron 6449.15 ms / OpenRouter 4341.44 ms	Infron 5771.49 ms / OpenRouter 3938.75 ms	Infron 0 / OpenRouter 8815
long	247	Infron 98.61% / OpenRouter 87.95%	Infron 98.79% / OpenRouter 90.28%	Infron 8556.78 ms / OpenRouter 6133.46 ms	Infron 7984.41 ms / OpenRouter 5738.76 ms	Infron 0 / OpenRouter 8829

Reasoning / Thinking Observation

This run does not explicitly disable or force reasoning/thinking. Reasoning tokens are treated as observed telemetry.

Routing mode	Infron reasoning tokens	OpenRouter reasoning tokens	Infron avg/request	OpenRouter avg/request
Throughput First	0	8247	0.0000	22.1694
Price First	0	6078	0.0000	15.9947

Routing mode	Infron reasoning tokens	OpenRouter reasoning tokens	Infron avg/request	OpenRouter avg/request
Latency First	0	6034	0.0000	15.9630
TTFT First	0	6130	0.0000	15.9635

Reproducibility References

Artifact	GitHub path	SHA256 / Notes
Summary	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/data/summary.json	534296f46e82f6a2e15221593b519fb7ff8c03e45fc719d0b7a75f7c91
Paired dataset CSV	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/data/benchmark_pairs.csv	f892a492f0d9857164dd1f6faf33205df393957cf851230c3cc695defb
Request-level dataset JSONL	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/data/benchmark_requests.jsonl	237b16fd9bc60c9fa7bc2c4e2365bf2b3bcfb1f88978b357c2cc955589
Excluded-record audit	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/data/records_excluded.json	ee6b2fc4533de02d659e3c2f21f24a8b1a587a5a99e6e8cbd7cea7186c
Benchmark runner source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/code/rerun_routing_sort_cache_cost_ab.py	a485c5116e99375e6b171d0328e5a423babd1ee6b206c72f54053a3a04
HTML report renderer source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/code/render_glm52_deepseek_style_report.py	e5bb0d32361d70ee688e0d0a0da3302a1e542e33e53876c905814eb5c2

Artifact	GitHub path	SHA256 / Notes
Test source	https://github.com/InfronAI/prompt-cache-bench/blob/main/tests/test_rerun_routing_sort_cache_cost_ab.py	Repository-level test source path
Reports directory	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/reports	Bilingual HTML / Markdown / PDF reports
Data directory	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/data	Pair data, request telemetry, summary, and excluded records
Manifest	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/metadata/manifest.json	File sizes and SHA-256 checksums

Dataset reference: `business_representative` built-in representative business templates; request-level export is `benchmark_requests.jsonl`.

Online HTML: Chinese https://infronai.github.io/prompt-cache-bench/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_glm-5-2_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers_2026-07-04.zh.html; English https://infronai.github.io/prompt-cache-bench/experiments/z-ai/glm-5.2/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_glm-5-2_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers_2026-07-04.en.html.